Quantifying Time Travel



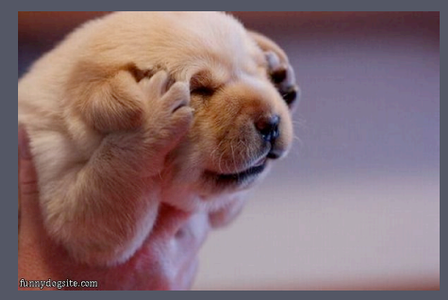
Dave Bacon
Department of Computer Science & Engineering
Department of Physics
University of Washington

At Christmas Time I Do Foundations



Physics => Computer Science

Today I will act like a theoretical computer scientist and try to think simply about "models of computing" or "information processing protocols." If you are a physicist please cover your ears if it gets to be too much.



Claim: "Models of computing" are a good way to free your mind of (important!) physical details, and could even lead to greater understanding of foundational issues (right!)

Theoretical CS

"If you think this Universe is bad, you should see some of the others."

— Philip K. Dick

Turing Machine

TM

Machine with finite set of states Q

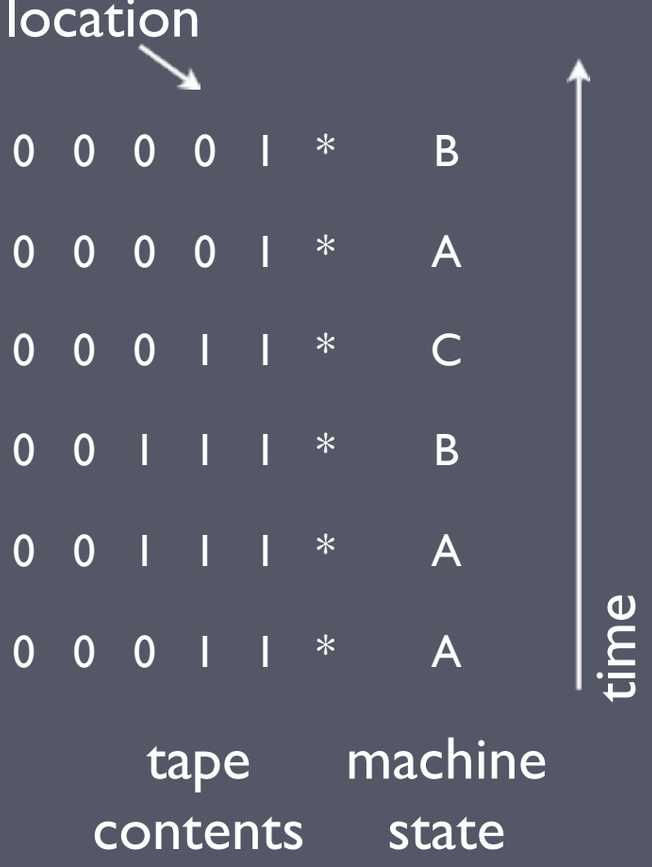
Tape with memory cells that can be filled with elements from some alphabet Σ

Transition function: Given symbol under machine + state, write symbol, update state, and move

$$\delta : Q \times \Sigma \to Q \times \Sigma \times \{L, R\}$$

Set of states where machine accepts input $A \subseteq S$

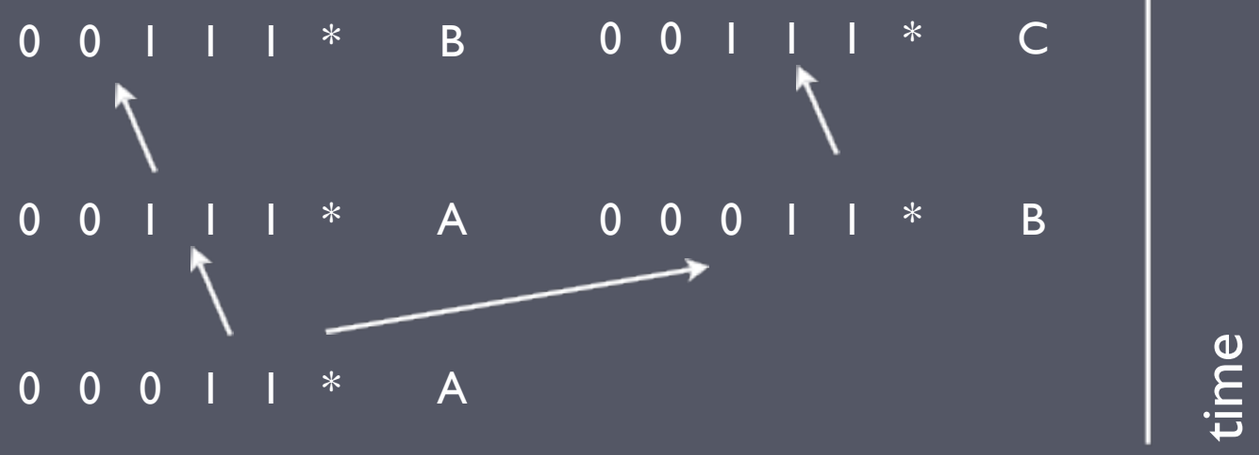
Start state $q_0 \in Q$



Non-deterministic Turing Machine

Transition function: Given symbol under machine + state, transition to one of a set of (write symbol, update state, and move)

$$\delta : Q \times \Sigma \to \mathcal{P}(Q \times \Sigma \times \{L, R\})$$



accept when one of the branches is in an accept state

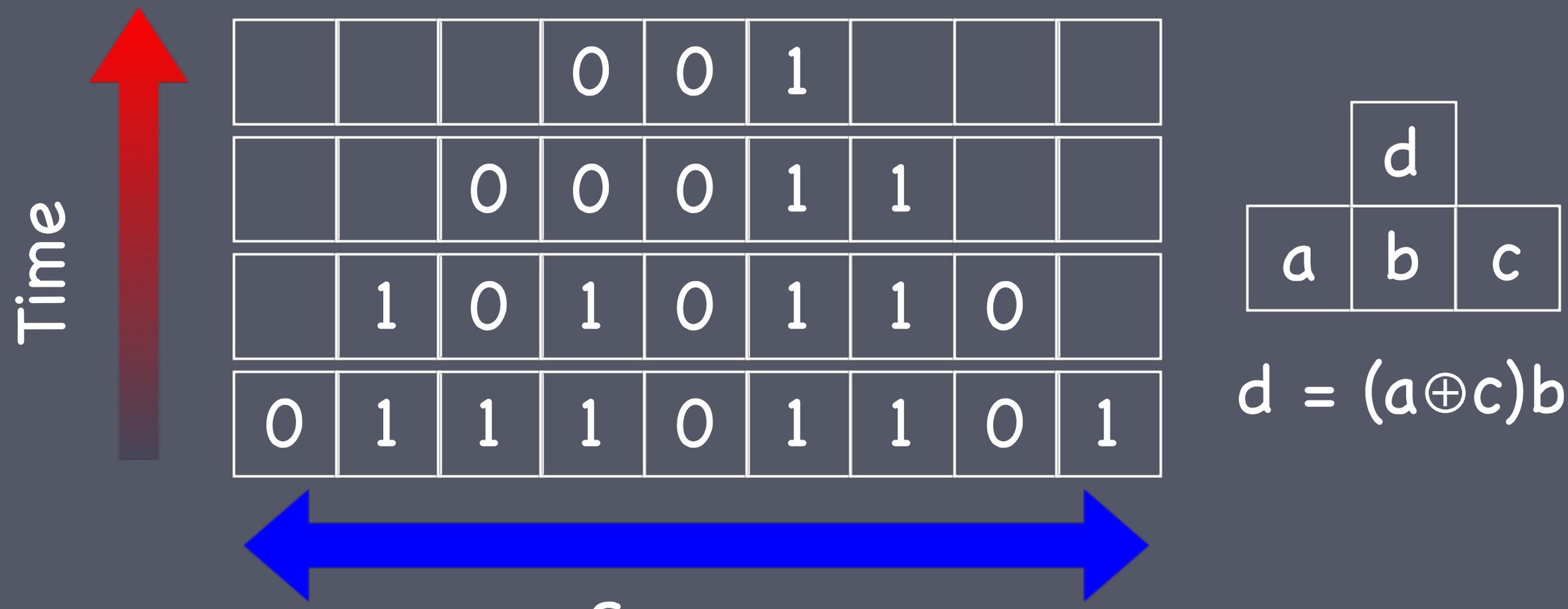
Imagine a universe where laws of physics are like this???

"Time is nature's way of keeping everything from happening at once.

Space is what prevents everything from happening to me."

- attributed to John Wheeler

Causal Structure



Very naive definition: which events can influence which other events

(substitute favorite alternative for word 'events' and 'influence')

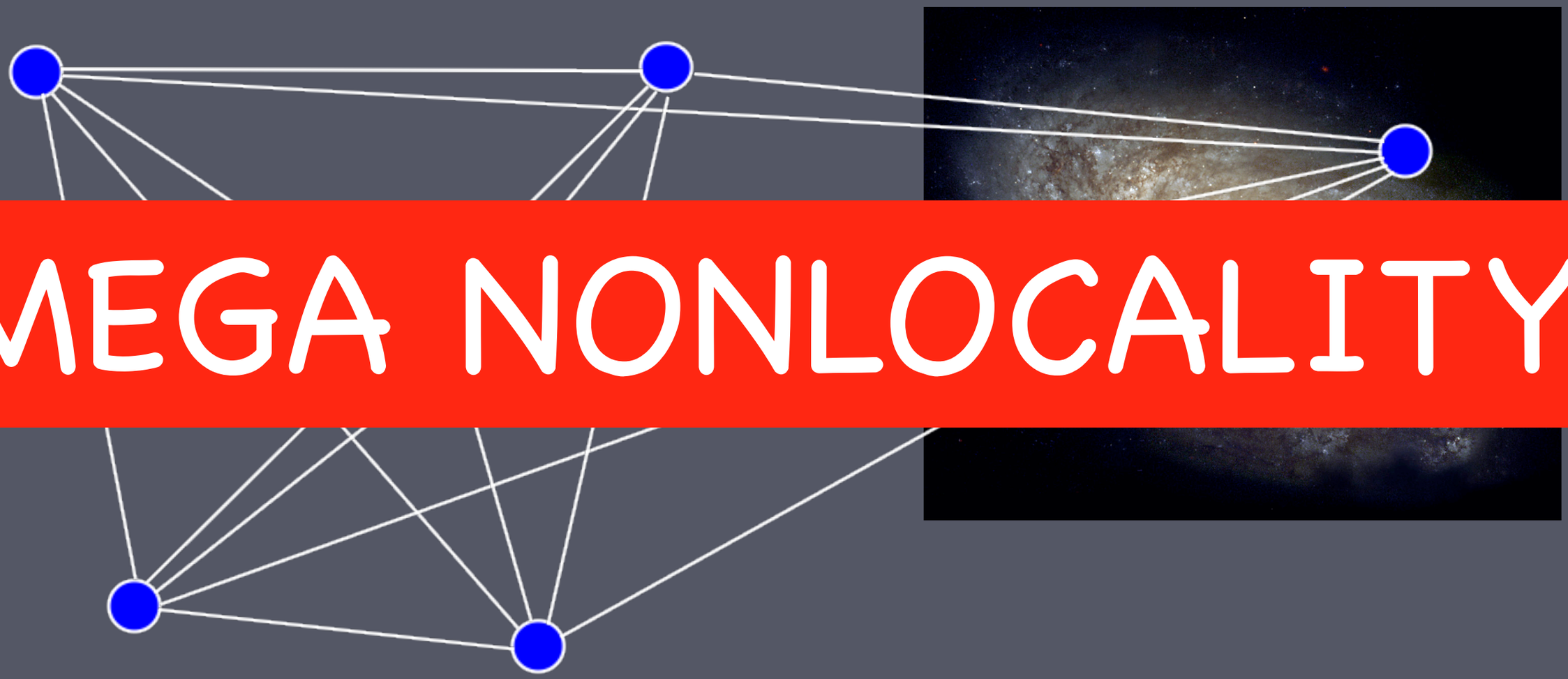
Bell's Theorem

There is no local hidden variable theory that can explain the correlations predicted by quantum theory.



"reasonable" causal structure + assumption of "local hidden variables" ≠ quantum theory

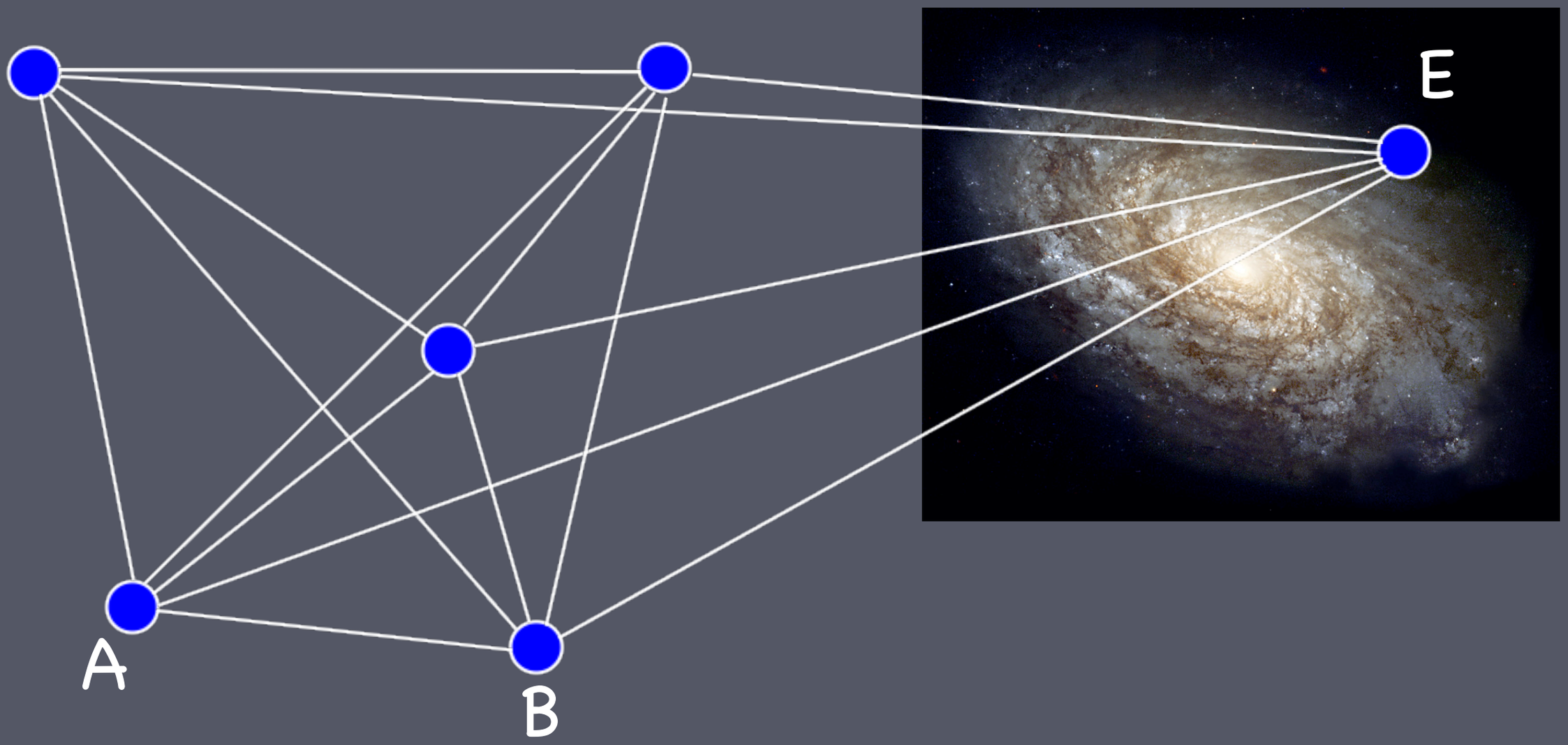
What If...



MEGA NONLOCALITY

....every subsystem could interact non-locally with every other subsystem in the universe?

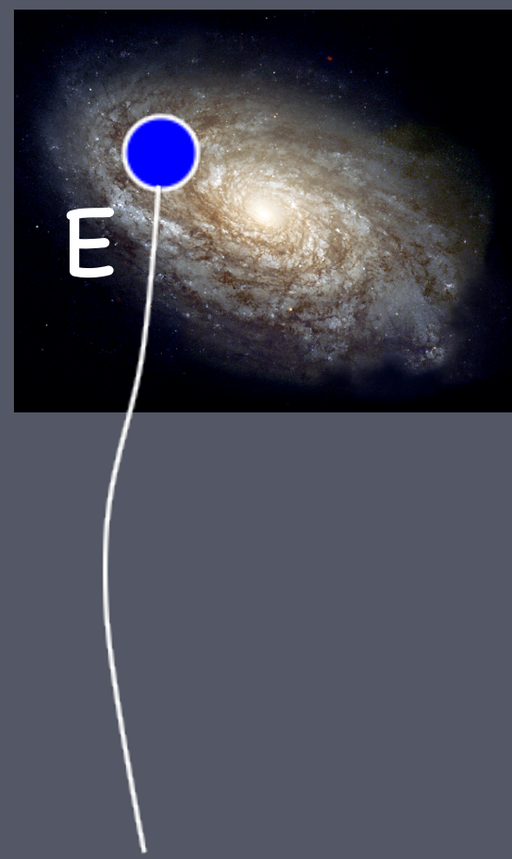
New Mystery



Why do A and B exhibit correlations, but A and E (B and E) do not?

One Answer

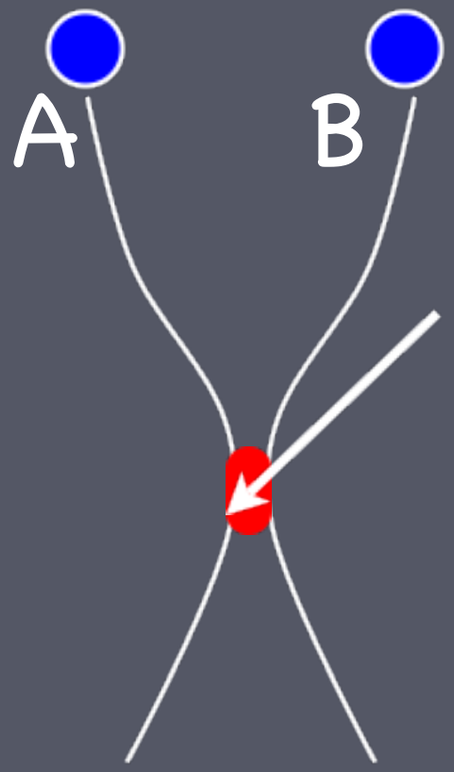




Correlations can exists between A and B because they interacted in the past.

A and B must be able to remember they have interacted in the past

Identity = Shared Key?



exchange some information that allows A and B to identify each other



Whatever information E transmits to the rest of the universe, it shouldn't destroy quantum correlations between A and B

Singlet Experiment

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$



$A \in \{+1, -1\}$





$\hat{a}$

$B \in \{+1, -1\}$







$\hat{b}$

two outcome
projective measurements

$$Pr[A, B|\hat{a}, \hat{b}] = \frac{1}{4}\left(1 - \hat{a}\hat{b}AB\right)$$

Singlet Protocol

1. Alice and Bob share two unit vectors, $\vec{\lambda_1}$ and $\vec{\lambda_2}$, uniformly distributed on the sphere

2. Alice outputs $-\mathrm{sgn}(\vec{a} \cdot \vec{\lambda_1})$

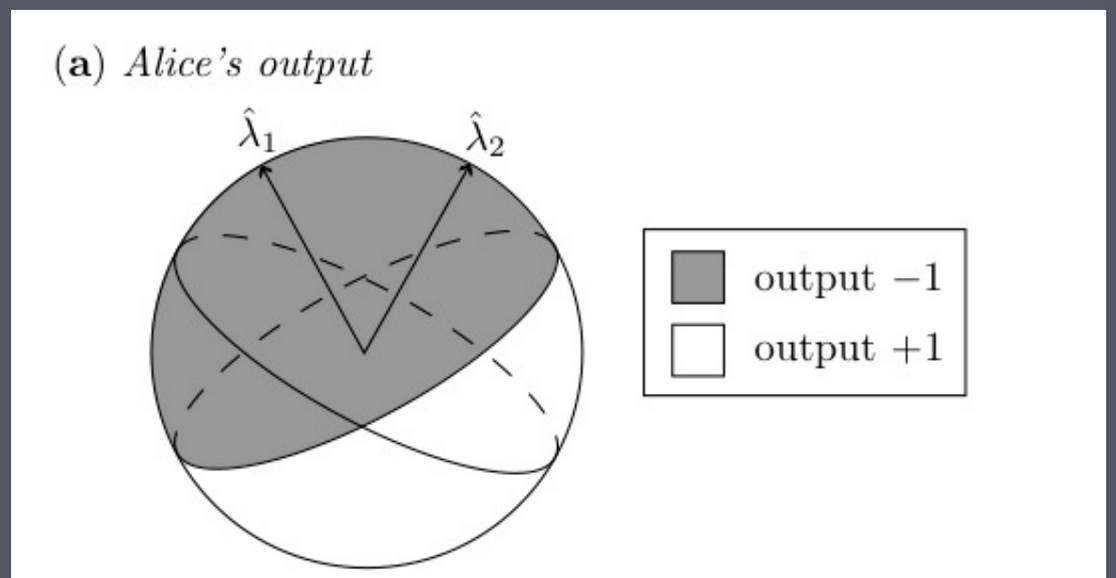
3. Alice sends one bit to Bob, $c = \mathrm{sgn}(\vec{a} \cdot \vec{\lambda_1})\mathrm{sgn}(\vec{a} \cdot \vec{\lambda_2})$

4. Bob outputs $\mathrm{sgn}[\vec{b} \cdot (\vec{\lambda_1} + c\vec{\lambda_2})]$

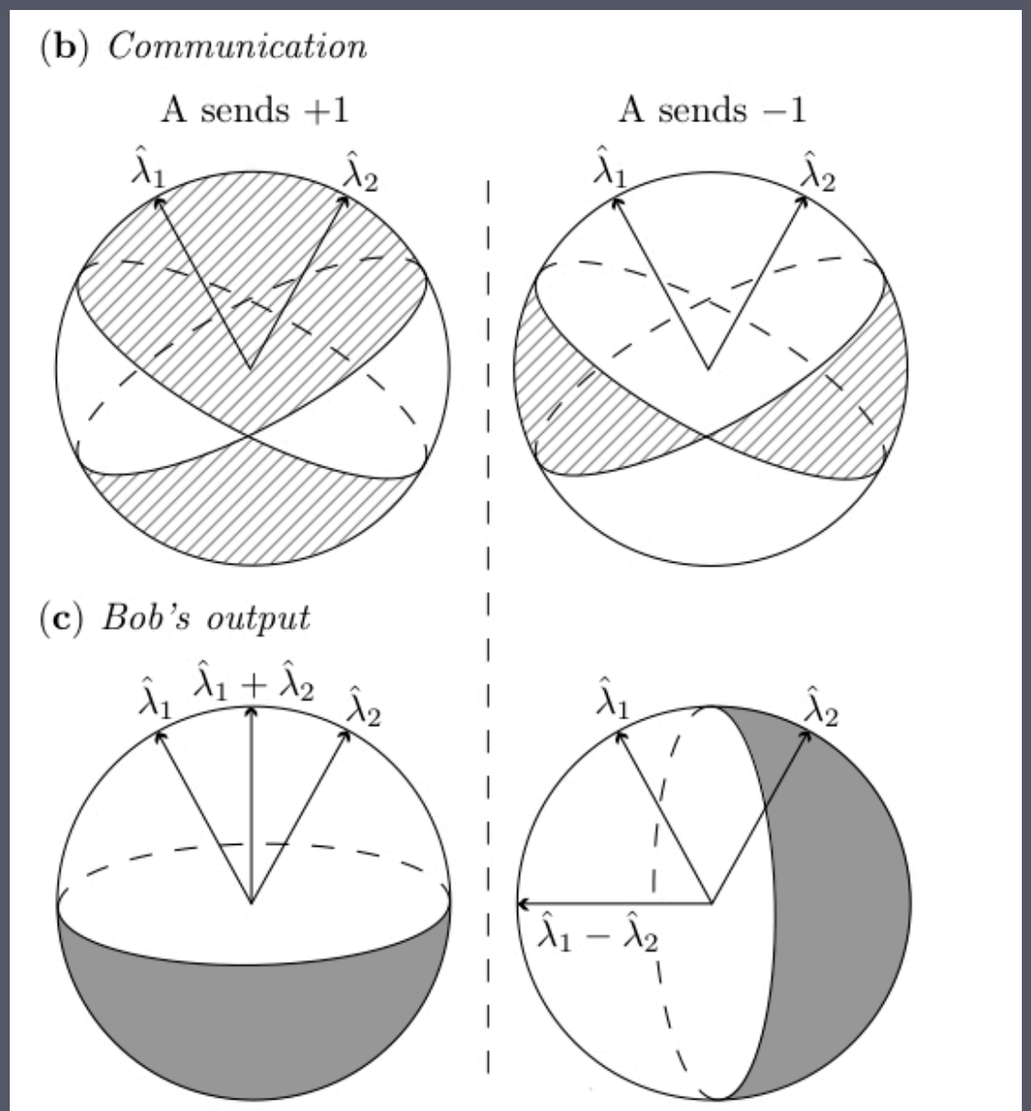
[Toner and Bacon, PRL, 2003]

Pictorially

$$c = \mathrm{sgn}(\vec{a} \cdot \vec{\lambda_1})\mathrm{sgn}(\vec{a} \cdot \vec{\lambda_2})$$



$$-\mathrm{sgn}(\vec{a} \cdot \vec{\lambda_1})$$



$$\mathrm{sgn}[\vec{b} \cdot (\vec{\lambda}_1 + c\vec{\lambda}_2)]$$

Proof Sketch

Both parties outputs change sign under

$$\vec{\lambda}_1 \leftrightarrow -\vec{\lambda}_1 \quad \vec{\lambda}_2 \leftrightarrow -\vec{\lambda}_2$$

thus $\langle A \rangle = \langle B \rangle = 0$

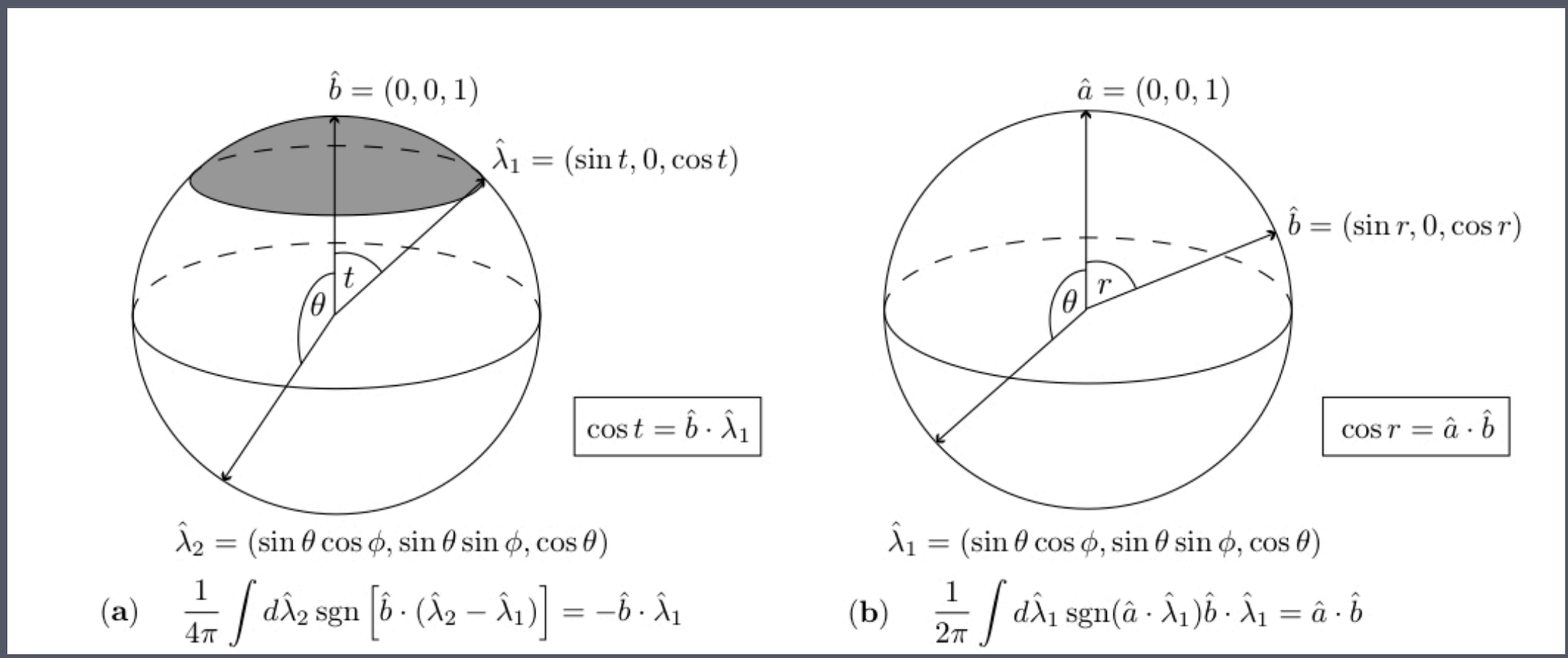
Plus an ugly calculation

$$\langle AB \rangle = E\left[-\mathrm{sgn}(\vec{a} \cdot \vec{\lambda}_1) \sum_{d=\pm 1} \frac{(1+cd)}{2} \mathrm{sgn}[\vec{b} \cdot (\vec{\lambda}_1 + d\vec{\lambda}_2)] \right]$$

$$E[x] = \frac{1}{(4\pi)^2} \int d\vec{\lambda}_1 \int d\vec{\lambda}_2 x$$

Babe Ruth

$$\langle AB \rangle = E\left[2\mathrm{sgn}(\vec{a}\cdot\vec{\lambda}_1)\mathrm{sgn}\left[\vec{b}\cdot(\vec{\lambda}_2-\vec{\lambda}_1)\right]\right] = -\vec{a}\cdot\vec{b}$$



(a) [K. H. Schatten, PRA 48 103 (1993)]
(b) [N. Gisin, B. Gisin, Phys. Lett. A 260, 323 (1999)]

LHV + Classical Communication

Quantify the amount of classical communication needed to augment a local hidden variable theory in order to reproduce quantum theory

[T. Maudlin 1992], [Brassard, Cleve, Tapp 1999], [Steiner 2000]

(Quantify information "on average", "worst case", or "with error".)

Open Question

Is it always possible to augment a local hidden variable theory with an amount of communication bounded by the dimension of the quantum system and reproduce quantum theory?

Progress

$A \in \{+1, -1\}$





ρ





$B \in \{+1, -1\}$

Bipartite quantum state

subsystems of dimension *d*

O_A

O_B

d x d observables with eigenvalues +1, -1

Simulate correlation: $E[AB] = \mathrm{Tr}[\rho(O_A \otimes O_B)]$

Progress

$A \in \{+1, -1\}$ $B \in \{+1, -1\}$




ρ




Simulate correlation:

O_A $E[AB] = \mathrm{Tr}[\rho(O_A \otimes O_B)]$ O_B

[Toner and Regev 2009]

This experiment can be simulated with local hidden variables plus two bits of communication (indpendent of d).

Embarrassment

Previous work. The problem of simulating quantum correlations was introduced independently by several authors, including [23, 29, 8]. It is also closely related to a communication complexity problem introduced by Kremer, Nisan, and Ron [20]. Early work concentrated on the special case of dimension $n = 3$, which turns out to correspond to the case of a shared EPR pair. The protocol in [8] solves this special case with 8 bits of communication; this was subsequently improved to just one bit [30]

(see also [10]). Up to now, the best known protocol for the general case of Problem 2 required $\lfloor n/2 \rfloor$ bits of one-way communication [4].

$n=d^2$

[3] S. Arora, E. Berger, E. Hazan, G. Kindler, and S. Safra, *On non-approximability for quadratic programs*, in Proceedings of the 46th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2005, pp. 206–215.

[4] D. Bacon and B. F. Toner, *How to simulate quantum correlations*, unpublished.

[5] J. Barrett, A. Kent, and S. Pironio, *Maximally nonlocal and monogamous quantum correlations*, Phys. Rev. Lett., 97 (2006), 170409.

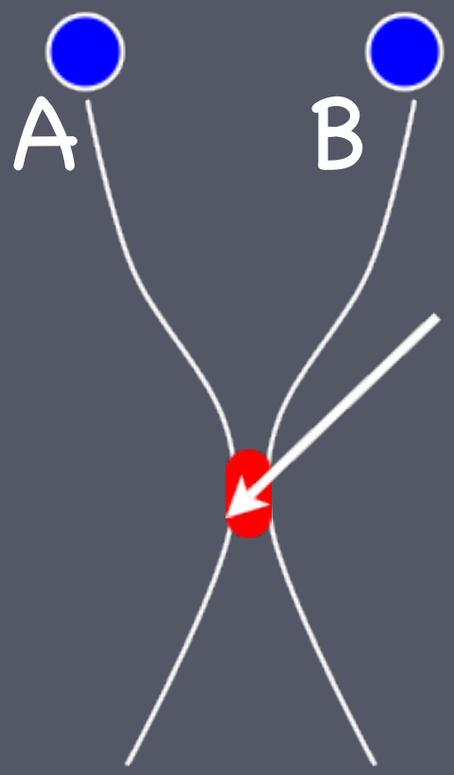
[6] J. S. Bell, *On the Einstein-Podolsky-Rosen paradox*, Physics, 1 (1964), pp. 195–200.

Open Question

Is it always possible to augment a local hidden variable theory with an amount of communication bounded by the dimension of the quantum system and reproduce quantum theory?

Dream: could we even boost this up to some sort of interpretation of quantum theory?

A Problem



exchange some information that allows A and B to identify each other



Whatever information E transmits to the rest of the universe, it shouldn't destroy quantum correlations between A and B

Singlet Protocol + Eve

1. Alice and Bob share two unit vectors, $\vec{\lambda_1}$ and $\vec{\lambda_2}$, uniformly distributed on the sphere

2. Alice outputs $-\mathrm{sgn}(\vec{a}\cdot\vec{\lambda_1})$

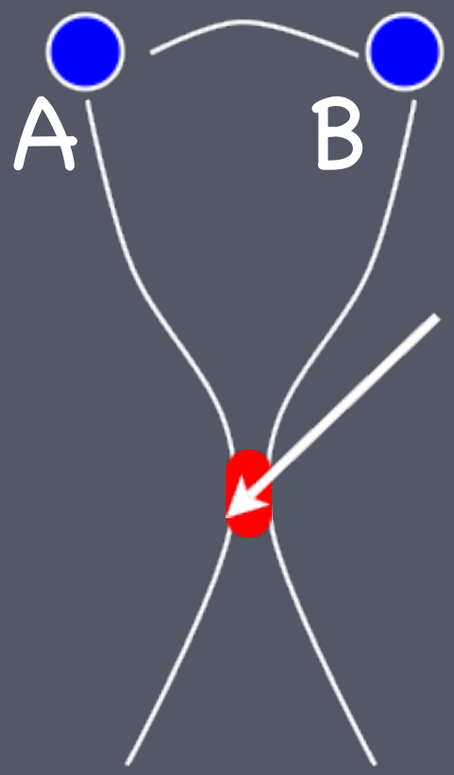
3. Alice sends one bit to Bob, $c=\mathrm{sgn}(\vec{a}\cdot\vec{\lambda_1})\mathrm{sgn}(\vec{a}\cdot\vec{\lambda_2})$

3a. Eve sends one bit to Bob, $e=\mathrm{sgn}(\vec{d}\cdot\vec{\sigma}_1)\mathrm{sgn}(\vec{d}\cdot\sigma_2)$

4. Bob outputs $\mathrm{sgn}[\vec{b}\cdot(\vec{\lambda}_1+ce\vec{\lambda}_2)]$

$$\langle AB\rangle = E\left[-\mathrm{sgn}(\vec{a}\cdot\vec{\lambda}_1)\sum_{d=\pm1}\frac{(1+ced)}{2}\mathrm{sgn}[\vec{b}\cdot(\vec{\lambda}_1+d\vec{\lambda}_2)]\right]=-\frac{1}{2}\vec{a}\cdot\vec{b}$$

A Problem



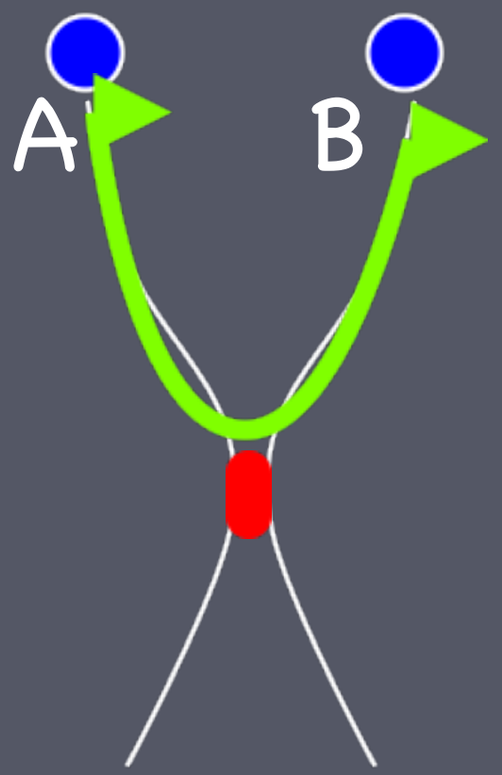
A and B must be able to "authenticate each other's identity"

exchange some information that allows A and B to identify each other



Whatever information E transmits to the rest of the universe, it shouldn't destroy quantum correlations between A and B

Is There a `Better' Solution?



"Transactional interpretation?"

Closed time like curves?

NEXT: A Complexity Theory approach to thinking about this question.

Closed Timelike Curves

VOLUME 61, NUMBER 13 PHYSICAL REVIEW LETTERS 26 SEPTEMBER 1988

Wormholes, Time Machines, and the Weak Energy Condition

Michael S. Morris, Kip S. Thorne, and Ulvi Yurtsever
Theoretical Astrophysics, California Institute of Technology, Pasadena, California 91125
(Received 21 June 1988)

Normally theoretical physicists ask, "What are the laws of physics?" and/or "What do those laws predict about the Universe?" ... We ask, instead, "What constraints do the laws of physics place on the activities of an arbitrarily advanced civilization?" ... whether the laws of physics permit an arbitrarily advanced civilization to construct and maintain wormholes. Morris, Thorne, Yurtsever PRL, 61, 1446 (1988)

Closed Timelike Curves

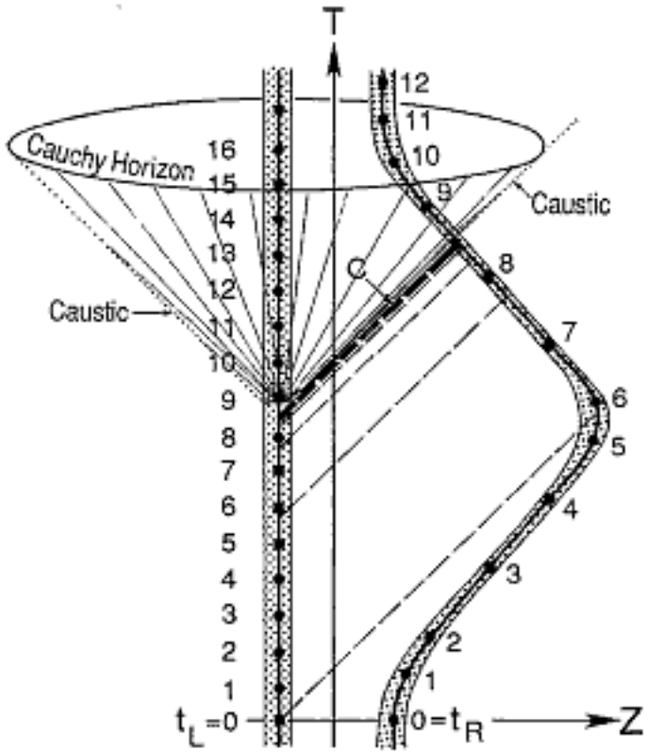


FIG. 2. Spacetime diagram for conversion of a wormhole into a time machine.

Use stable wormhole plus time dilation to make a time machine

but is this stable?

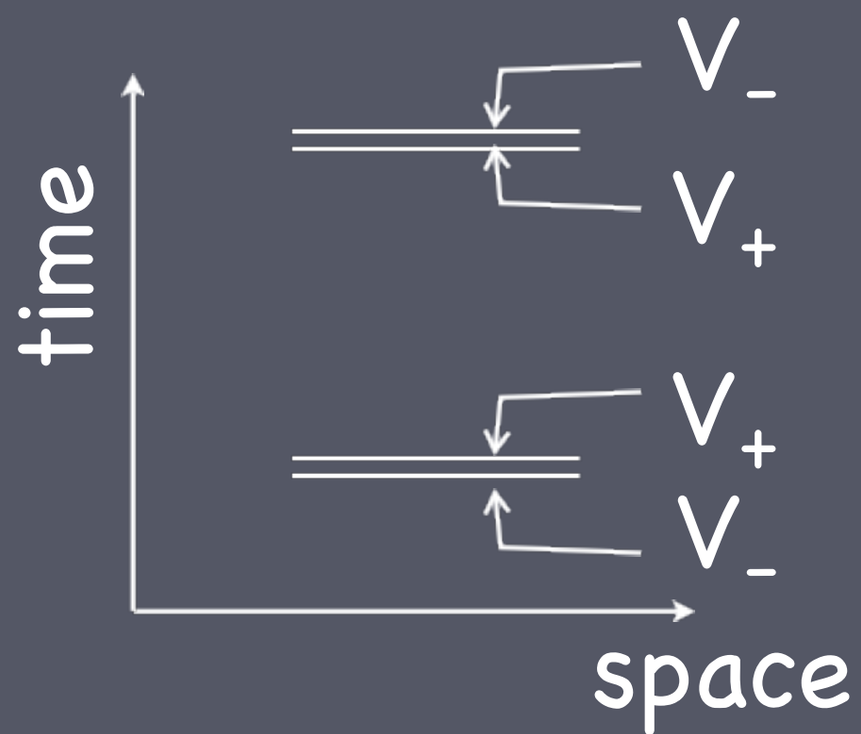
PHYSICAL REVIEW D VOLUME 46, NUMBER 2 15 JULY 1992

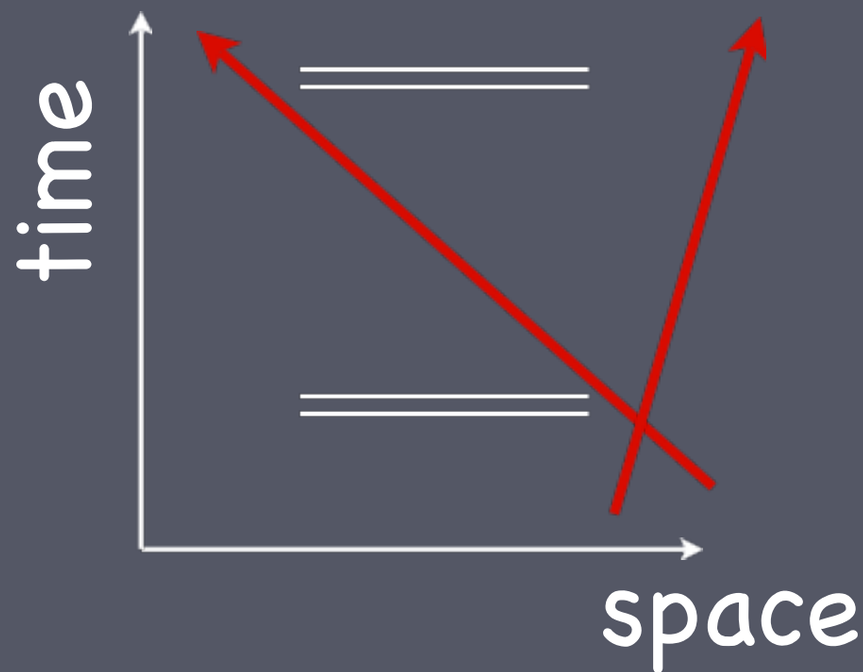
Chronology protection conjecture

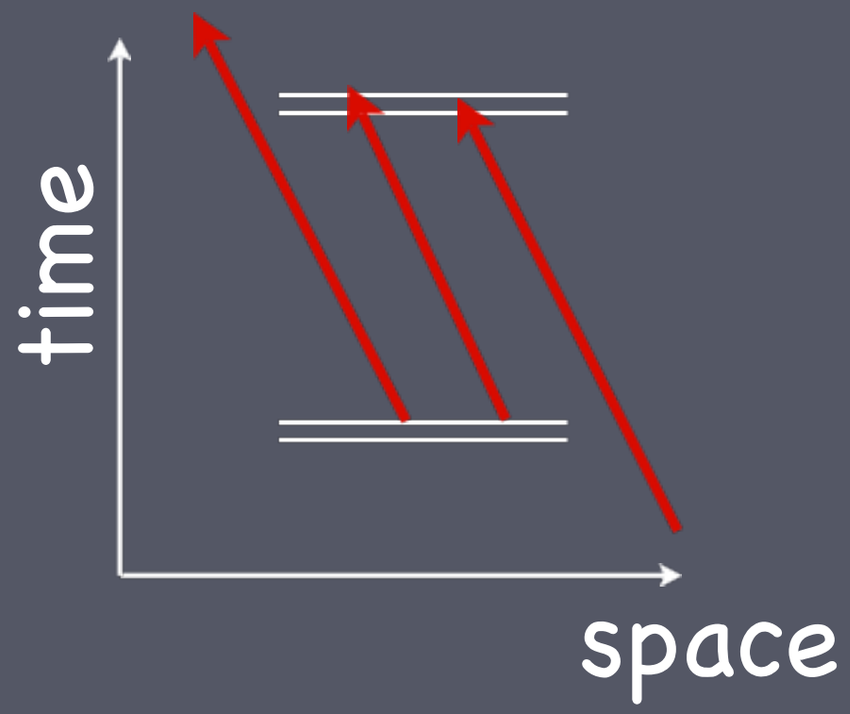
S. W. Hawking
Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Silver Street, Cambridge CB3 9EW, United Kingdom
(Received 23 September 1991)

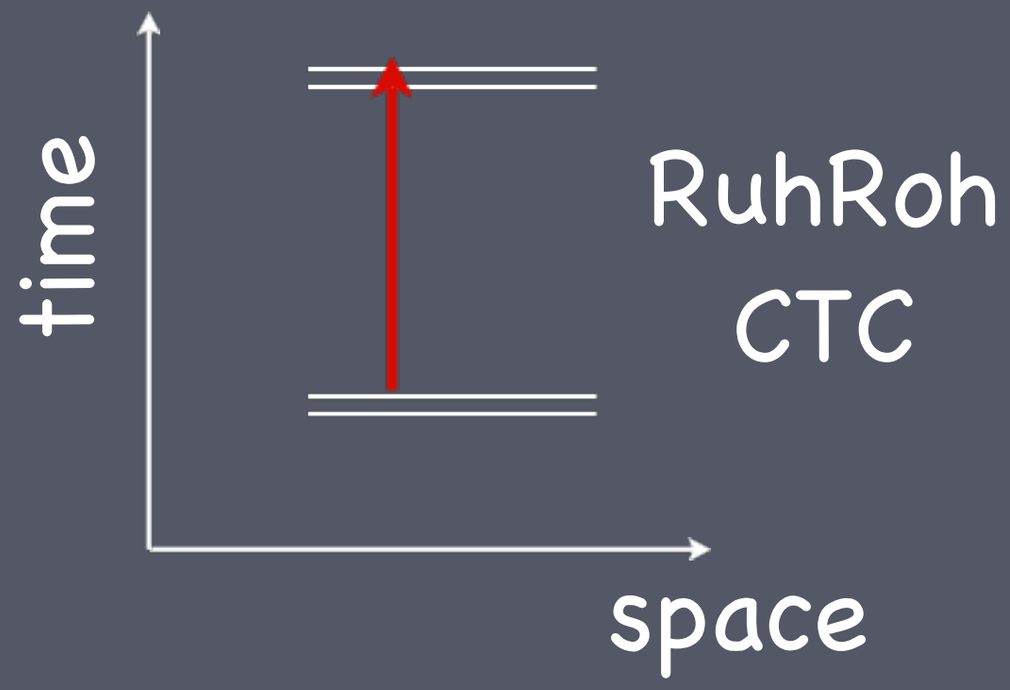
Closed Timelike Curves

[Politzer, PRD 49, 3981 (1994)]

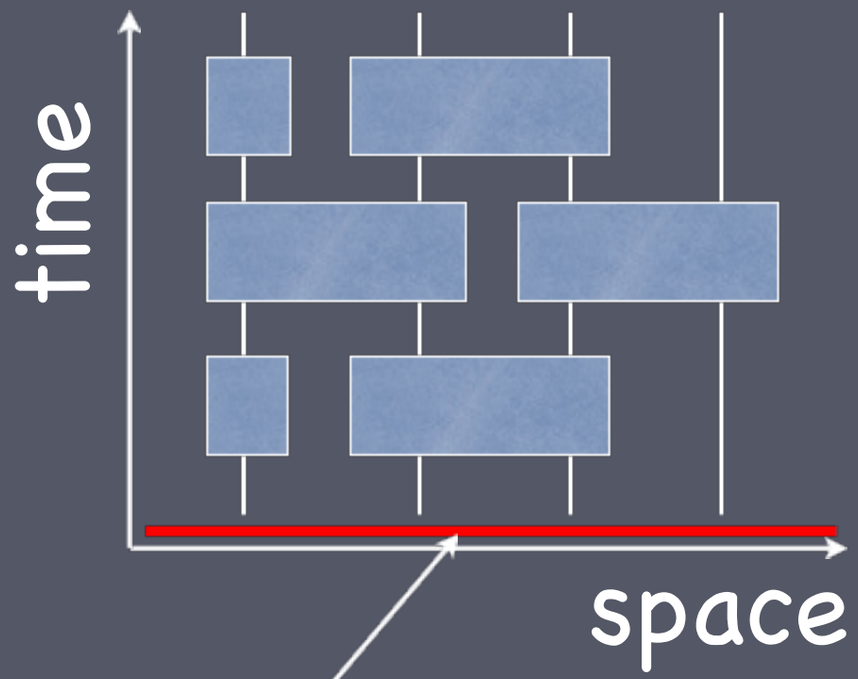




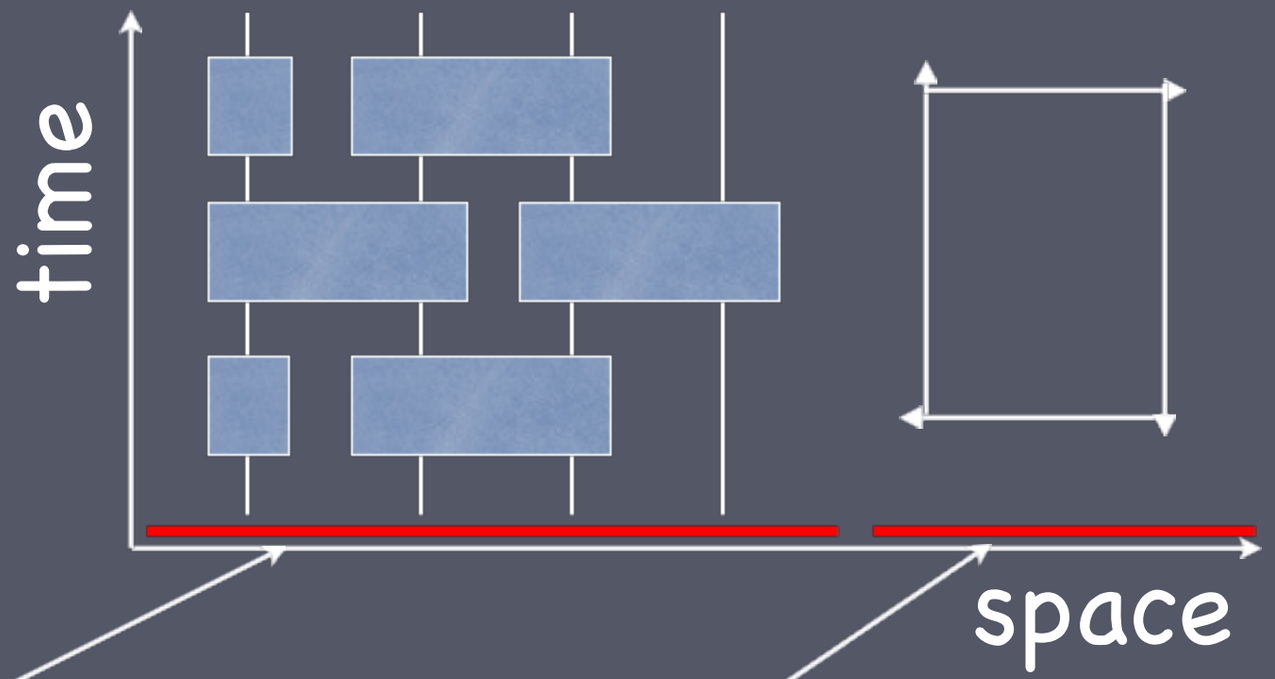




Qubits in Spacetime



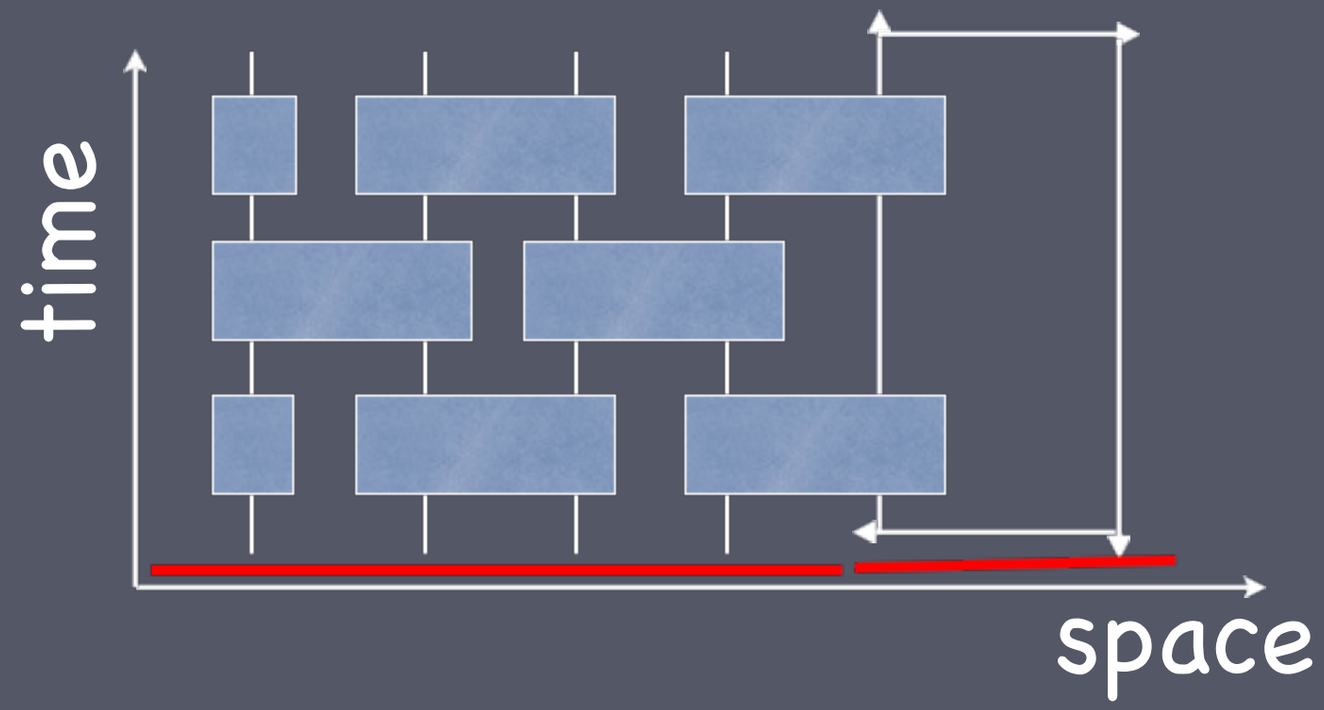
initial conditions specifying input to circuit and gates



initial conditions specifying creation of CTC

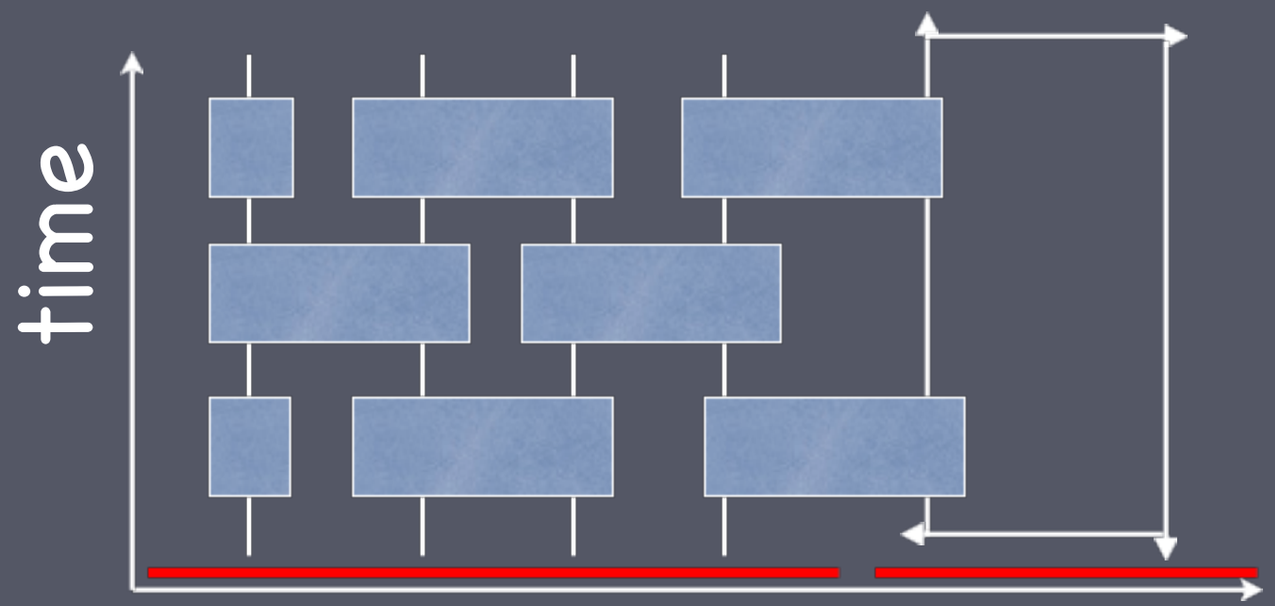
Complexity: count creating of CTC as extra resource in circuit.

Qubits in Spacetime



Allow CTC qubits to interact with chronology respecting qubits

Consistency

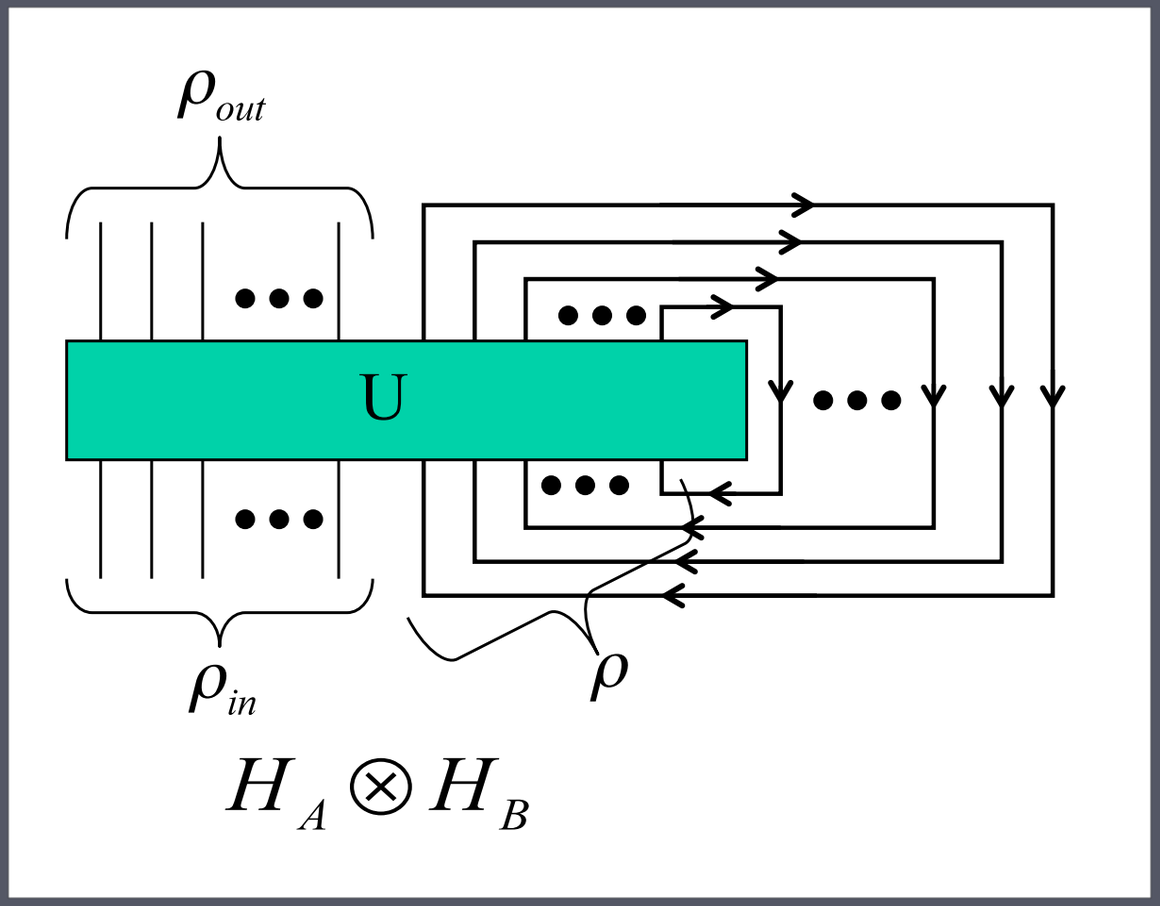


to (Quantum Theory) add (Consistency) in presence of (CTCs)

Principle of self-consistency: "the only solutions to the laws of physics that can occur locally are those which are globally self-consistent."

I.D. Novikov, Evolution of the Universe (1983)
Friedman et al. "Cauchy problem in spacetimes with closed timelike curves" PRD 42 1915 (1990)
D. Deutsch "Quantum mechanics near closed timelike lines" PRD 44 3197 (1991)

Consistency?



consistency:

$$\rho = \mathrm{Tr}_A[U(\rho_{in} \otimes \rho)U^\dagger]$$

input-ouput:

$$\rho_{out} = \mathrm{Tr}_B[U(\rho_{in} \otimes \rho)U^\dagger]$$

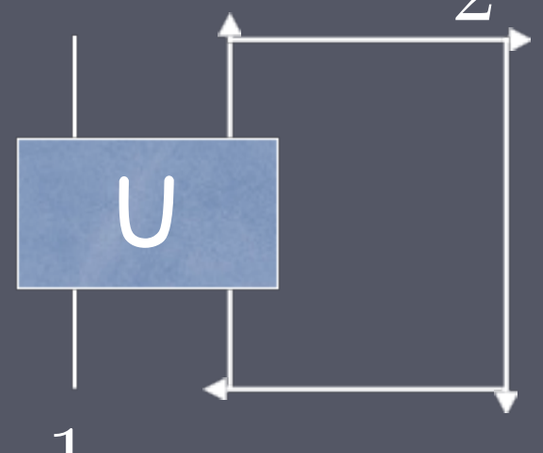
non-linear in ρ_{in}

Questions: Always a consistent evolution? YES
Always a unique consistent evolution? NO

[Deutsch 1991]

Example

ρ_{out} $\rho = \frac{1}{2}(I + \vec{m} \cdot \vec{\sigma})$



$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$\rho_{in} = \frac{1}{2}(I + \vec{n} \cdot \vec{\sigma})$

Consistency: $m_x = n_x m_z \quad m_y = n_y m_z \quad m_z = n_z$

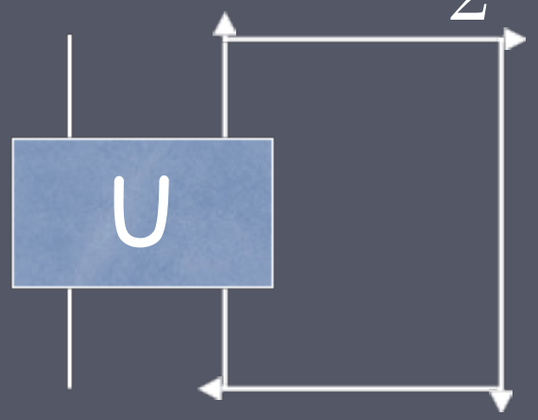
(single consistent CTC state)

Output is unambiguous:

$$\rho_{out} = \frac{1}{2}(I + n_z^2 n_x \sigma_X + n_z^2 n_y \sigma_Y + n_z \sigma_Z)$$

Example

ρ_{out} $\rho = \frac{1}{2}(I + \vec{m} \cdot \vec{\sigma})$



$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$\rho_{in} = \frac{1}{2}(I + \vec{n} \cdot \vec{\sigma})$

Consistency: $m_x = n_x m_x \quad m_y = n_y m_x \quad m_z = n_z$

Output is unambiguous if $n_x \neq 1$

$$\rho_{out} = \frac{1}{2}(1 + n_z^2 \sigma_Z)$$

Output is ambiguous if $n_x = 1$

$$\rho_{out} = \frac{1}{2}(1 + m_x \sigma_X)$$

Ambiguity

Deutsch's proposed solution [Deutsch 1991]:

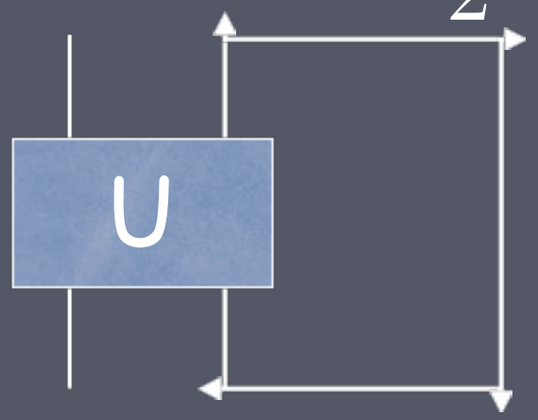
$$\max_{\rho|\rho \text{ is consistent}}(-\mathrm{Tr}[\rho \ln \rho])$$

Another proposed solution [Bacon 2004]: ambiguity could be resolved by initial conditions of CTC generating region.

If one considers open quantum systems one can argue for unique fixed point via the same arguments one tries to use to guarantee equilibrium [Ralph and Meyers 2010]

Blowing Up Probabilities

ρ_{out} $\rho = \frac{1}{2}(I + \vec{m} \cdot \vec{\sigma})$



$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$\rho_{in} = \frac{1}{2}(I + \vec{n} \cdot \vec{\sigma})$

Output is unambiguous if $n_x \neq 1$

$$\frac{1}{2}(I + n_z \sigma_Z) \rightarrow \frac{1}{2}(I + n_z^2 \sigma_Z)$$

By repeating n times we can distinguish $n_z = 1$ and $n_z = 1 - O(c^{-n})$

Back to CS Land

What is the computational power of qubits with CTCs?

Cook-Levin Theorem



Theorem: the **Boolean Satisfiability Problem** is **NP-complete**

Boolean formula (CNF):

Boolean variables $\{x_1, \ldots, x_n\}$

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_k$$

$$C_i = (t_{i1} \vee t_{i2} \vee \cdots \vee t_{il_i})$$

$$t_{ij} \in \{x_1, \ldots, x_n, \bar{x}_1, \ldots, \bar{x}_n\}$$

Boolean Satisfiability: does there exist an assignment of $\{T, F\}$ to the boolean variables such that $\phi = T$?

Cook-Levin Theorem



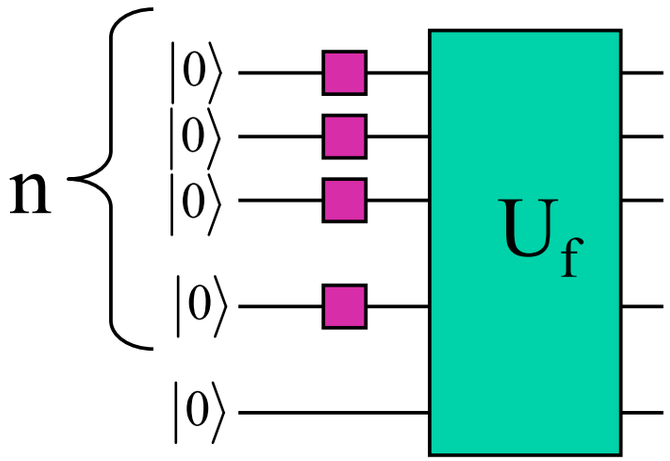
Theorem: the **Boolean Satisfiability Problem** is **NP-complete**

NP: the class of languages such that the language is accepted by a non-deterministic Turing machine

Language: A set of strings from the TMs alphabet

NP-complete: In NP and any problem in NP can be converted to the problem in polynomial time.

Power of CTCs



$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle|f(x)\rangle$$

$$\rho_0 = \begin{bmatrix} 1-\frac{s}{2^n} & 0 \\ 0 & \frac{s}{2^n} \end{bmatrix}$$

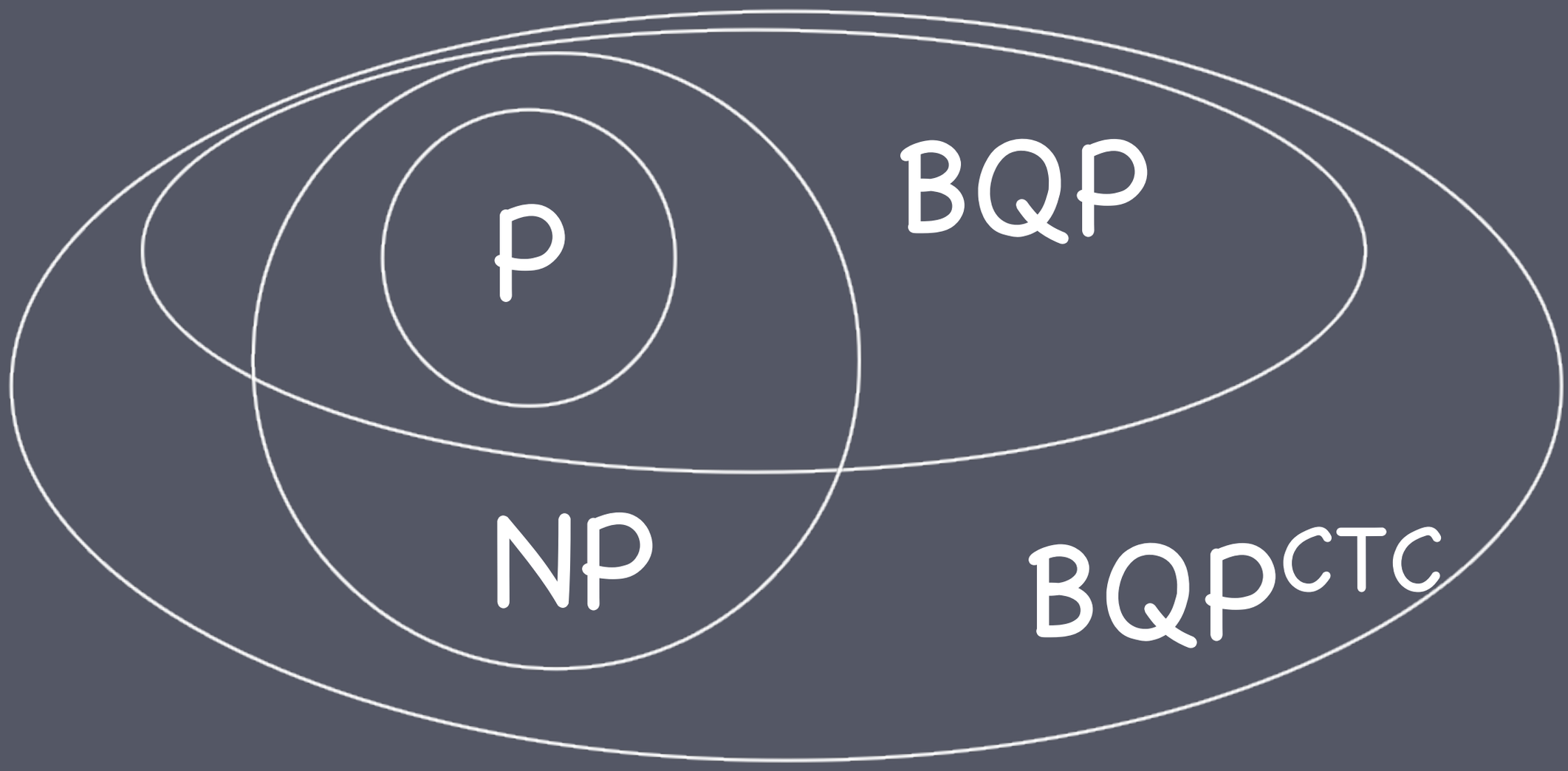
s is number of x's such that $f(x)=1$

Then repeatedly apply: $\frac{1}{2}(I + n_z\sigma_Z) \rightarrow \frac{1}{2}(I + n_z^2\sigma_Z)$

Can solve NP-complete problems in polynomial time

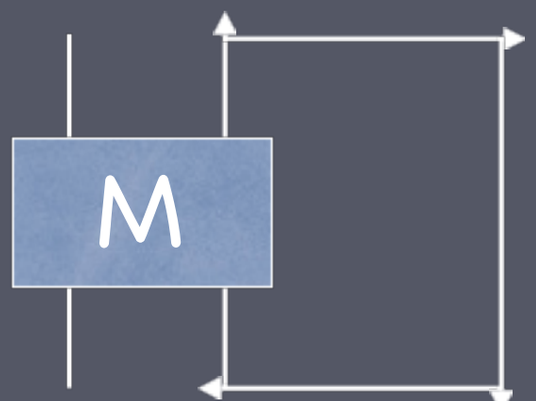
[Brun 2002, Bacon 2003]

Power of Deutsch CTCs [2003]



P = problems solvable in poly time on classical TM
BQP = problems solvable in poly time on a quatum TM

Anything You Can Do

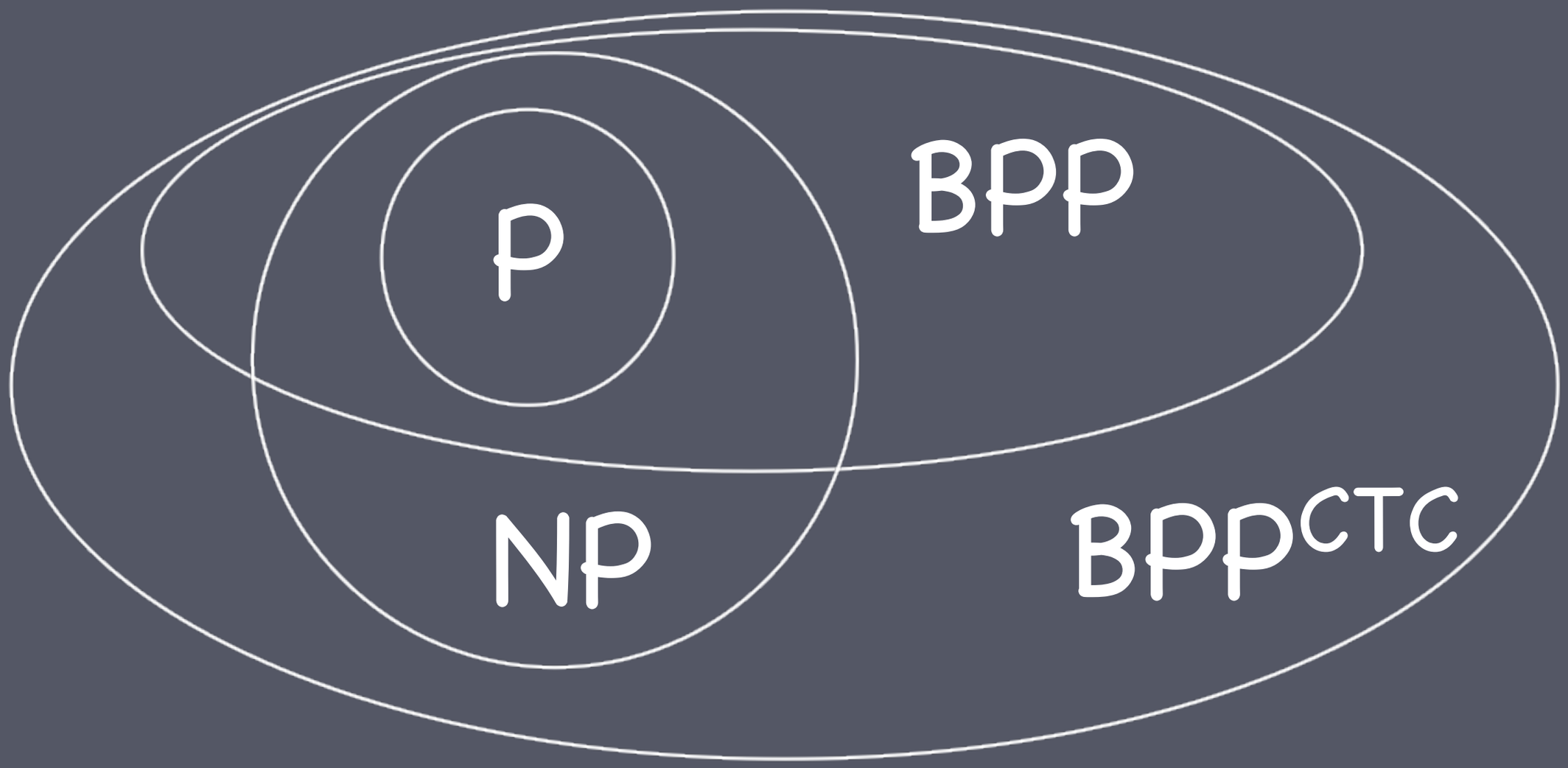


Replace quantum by classical
probabilistic computation
density matrix => probability distribution,
unitary matrix => stochastic matrix

$$Pr(0) \rightarrow Pr(0)^2$$

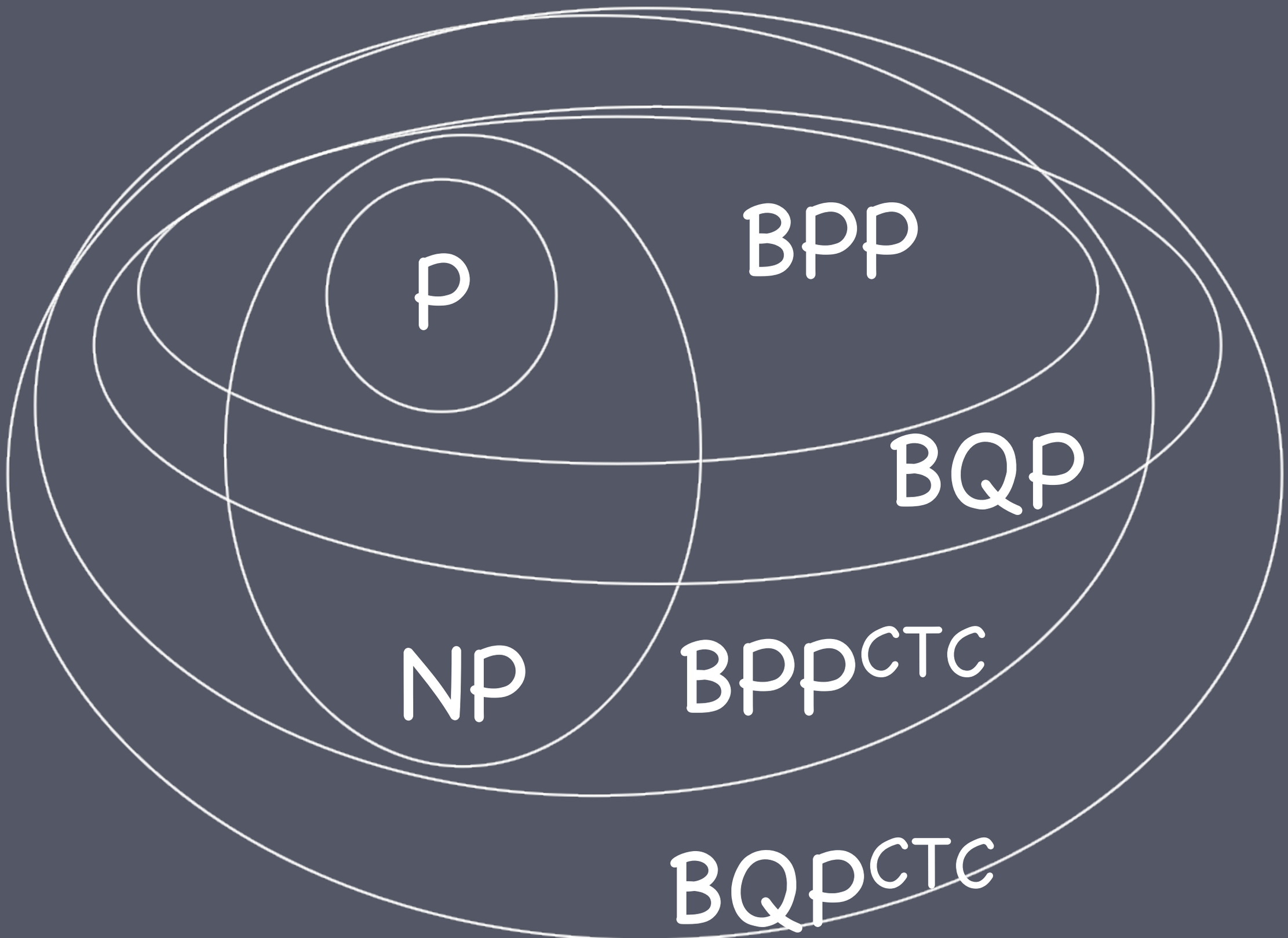
and we can similarly amplify probabilities

Power of CTCs [2003]



P = problems solvable in poly time on classical TM
BPP = problems solvable in poly time on aprobabilistic TM

Power of Classical CTCs



Is BQP in BPP^{CTC}

Translation: is there an "efficient" simulation of quantum theory using classical computation and closed timelike curves?

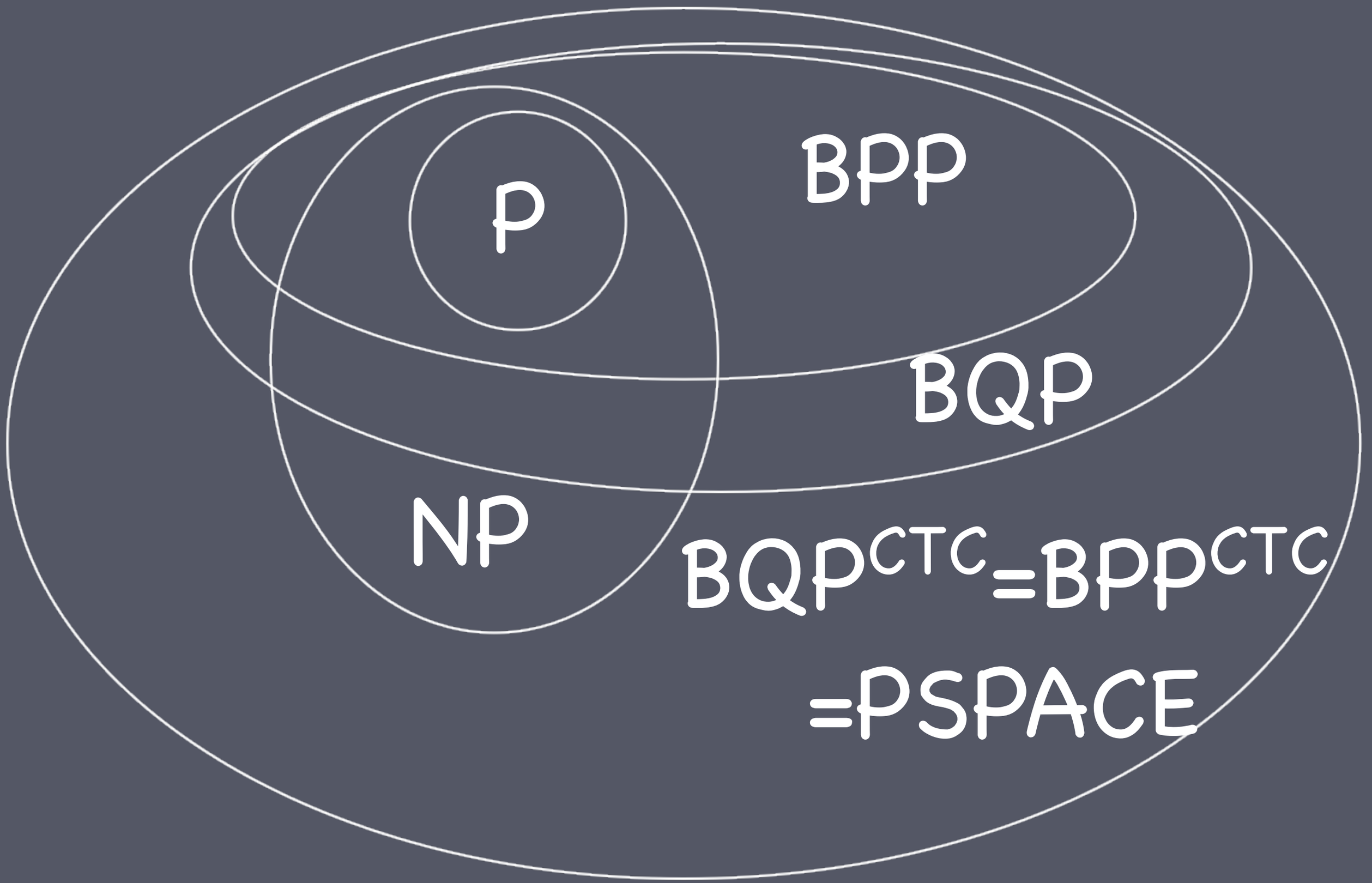
Breakthrough

$BPP^{CTC} = BQP^{CTC} = PSPACE$

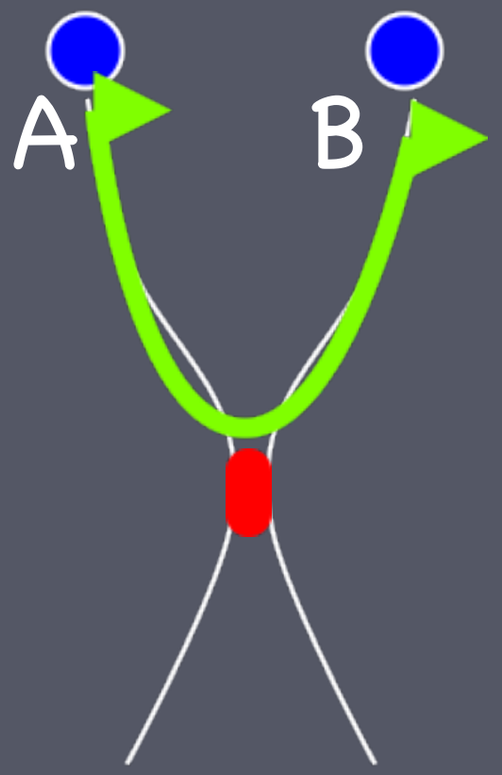
[Aaronson and Watrous 2008]

PSPACE = problems that can be solved on a Turing Machine using polynomial space (unlimited time)

Power of Classical CTCs



Is There a `Better' Solution?



There exists an efficient way to simulate quantum theory using classical probabilistic evolutions plus closed timelike curves

But can we do this in a way that respects the interaction network of the systems?

Trouble?

Non linear theories lead to all sorts of craziness (faster than light transmission) and most interesting the linearity trap [Bennett et al 2009]

$$|x\rangle|\phi_x\rangle \rightarrow |x\rangle|x\rangle$$

but

$$\sum_x p_x |x\rangle\langle x| \otimes |\phi_x\rangle\langle\phi_x| \nrightarrow \sum_x p_x |x\rangle\langle x| \otimes |x\rangle\langle x|$$

Are there restrictions where this linearity trap are hidden? This must be true for simulation of BQP.

Conclusion?

Classical local hidden variables + 1 bit can simulate projective measurements on singlet

Can local hidden variables + c bits simulate measurements on d dimensional systems with c=O(d)?

This seems to have problems as an interpretation of QM

Can CTC be used to efficiently and locally simulate QM?